Design

Architecture

main.py provides the command line. src/calculations.py contains side-effect-free Decimal formulas. src/engine.py validates CSV/JSON, writes source records, closes disputes, builds weekly statements, and exports reports. database/database.py opens SQLite and applies database/schema.sql on each connection. The existing database file is database/settlements.sqlite3.

Processing flow

- Register restaurant profiles.
- Import orders (including optional item rows and dispute status). A SHA-256 history entry makes identical file imports idempotent.
- Inspect `order-breakup` before settlement.
- Open disputes remain unassigned. Closing one records the timestamp; settlement selects the week containing that close timestamp.
- Weekly settlement selects unassigned eligible orders, calculates one statement, records per-order audit rows, then locks those orders to its unique statement.
- Reports export persisted order, settlement, and audit tables as CSV or JSON.

Money and reporting

All money values are Decimal text in SQLite and quantized to two decimal places with ROUND_HALF_UP. TDS GMV is iterated in chronological order without converting to binary floating point. Commission uses gross less restaurant-funded discount; packaging is separate. A sub-Rs 100 order pays the fixed platform fee instead of commission. Section 9(5) GST is displayed as a deduction because the platform withholds and remits it for the restaurant. Delivery fees are totalled for visibility but excluded from payout pending a specified recipient/funding rule; see the README for assumptions.

Inputs

Restaurant columns:
restaurant_id,name,plan,onboarding_date,rating,packaging_charge,weekly_ad_budget (optional status).
Order columns: order_id,restaurant_id,order_time,items_gross plus optional discounts, packaging, distance, mode, outcome, refund fields, dispute fields, cancel seconds, accepted flag, delivery fee, platform fee, rating. JSON order objects may add items:[{name,quantity,unit_price}].

Known scope choices

Rates and GST bases follow the supplied brief. Refund/cancellation customer funding is shown in customer_refund; it is deducted in the payout calculation. Delivery and platform fee values are retained in the order ledger because the brief does not define their formulas or recipient. Ratings are collected per order; the prior week average is used for next week's commission penalty. See decision_log.md for exact interpretations.